Advanced AI Assignment

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Branch: CSE (AIML)

Batch: A2

TASK FOR MSE

Task 1: Analytical comparison: **A* vs Greedy Best-First Search**

	A*	Greedy Best-First
Formula	uses f(n)=g(n)+h(n)	uses f(n)=h(n)
Completeness	Complete on finite graphs with non-negative edge costs. Will find a solution if one exists.	Not guaranteed complete; may get stuck or loop without extra checks.
Optimality	Optimal if heuristic is admissible (and consistent for graph search).	Not optimal; may find suboptimal paths
Time Complexity	Worst-case exponential: O(b^d), depends on branching factor (b) and solution depth (d); sensitive to heuristic quality.	Typically faster than A*, explores fewer nodes, but worst-case still high; heavily depends on heuristic
Space Complexity	High — stores open/closed sets; worst-case O(b^d).	Usually lower than A*, but can be high in worst-case scenarios.
Domain of Application	Pathfinding, planning, robotics, games — situations where optimal path matters.	Quick approximate search, large search spaces, interactive applications where speed is prioritized over optimality.
Example Scenario	Warehouse robot minimizing energy/battery use — optimal routing required.	Exploratory drone navigating unknown space — speed more important than optimal path.
Recommended Choice	A* — ensures optimal and safe routes.	Greedy — faster, but may miss optimal solutions.

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Task 2: My Thoughts on Responsible AI in Healthcare

Let's think about AI in healthcare, like a tool that helps doctors read X-rays or spot diseases.

The Big Worries

1. **Bias & Fairness**
    - This is a huge one. If an AI is only trained on data from one group of people, it might fail to correctly diagnose everyone else.
    - **The Risk:** This could lead to genuinely unequal care, where the AI works better for some people than for others based on their race, gender, or age.
2. **Data Privacy & Confidentiality**
    - Medical records are incredibly private.
    - **The Risk:** If this sensitive data leaks or is misused, it's a massive breach of trust that can seriously harm patients and make people afraid to use these helpful new tools.

How We Can Tackle These Problems

- **To Fight Bias:**
    - We have to build better datasets from the start, making sure they represent all kinds of people.
    - We can use special training techniques that force the AI to be fair.
    - We need to constantly check its performance across different groups, not just on average.
    - Crucially, keep a "human in the loop"—a doctor who reviews the AI's suggestions, especially for tricky cases.
- **To Protect Privacy:**

- Use smart techniques like **Federated Learning**, where the AI trains on data at different hospitals *without* the raw data ever leaving.
- We can also use **Differential Privacy**, which adds "noise" to the data so you can't trace it back to a single person.
- This has to be backed by strong rules: strict encryption, access controls, and audits, all following laws like HIPAA or other national data acts.
- Be open with patients about how their data is used and only collect what's absolutely necessary.

My 100-Word Take

To keep AI trustworthy, especially in health, I think it's a mix of good tech and good rules. As engineers, we have to build diverse datasets and use privacy-first methods like federated learning. But tech isn't enough. We need to be able to explain *why* an AI made a decision and always have a human expert in charge. This must be backed by strong policies, clear patient consent, and real accountability. The only way this works is if engineers, doctors, ethicists, and patients all build it together.

Task 3: How Governments are Handling AI (India & a Global View)

Key AI Moves in India

- **NITI Aayog — National Strategy for AI (NSAI)**
    - Think of this as India's national game plan. It’s about using AI for the public good in big areas like healthcare, farming, and education, and setting the standards for how to do it right.
- **INDIAai (National AI Portal)**
    - This is like a public library and community center for AI. It’s a one-stop-shop for researchers, students, and businesses to find resources, share ideas, and build an AI community.
- **Responsible AI for Youth / YUVAi**
    - This is all about teaching kids (in grades 8-12) what AI is, how it works, and how to use it responsibly. It's building AI literacy from the ground up.
- **Digital Personal Data Protection (DPDP) Act, 2023**
    - This is the big new rulebook. It sets the legal foundation for how companies can collect and use digital personal data, balancing people's right to privacy with the need for innovation.

A Global Example: The EU

- **EU AI Act**
    - The EU is taking a "safety-first" approach. They're creating a rulebook that sorts AI systems by risk.
    - Things like AI-powered social scoring are banned (unacceptable risk).

- Things like medical diagnostic tools are "high-risk" and face strict rules on transparency, accuracy, and human oversight before they can be used.

How Do They Compare? (India vs. EU)

- **Their Main Goals**
  - **India:** The focus is on *growth and inclusion*. The motto is "AI for All." The goal is to boost the economy, solve big social challenges, and build up a skilled workforce, all while putting new data-privacy rules in place.
  - **EU:** The focus is primarily on *rights and safety*. The goal is to protect citizens' fundamental rights from the risks of AI, creating a trustworthy and accountable market.
- **Their Real-World Impact**
  - **India:** You see this spurring new government programs, skill-building for students (YUVAi), and a growing startup scene, all now guided by the new DPDP data law.
  - **EU:** This is setting a global standard for regulation. Companies everywhere who want to operate in the EU have to follow these rules, which means higher compliance costs but stronger, legally-enforced protection for people.

My Quick Reflection

I think India's approach of mixing innovation with inclusion is smart. Using AI to help in farming or healthcare and teaching kids AI skills at the same time creates real social value. As the new data laws (like DPDP) and institutions (like INDIAai) get stronger, this combination will be key. To really lead, I believe India needs to keep pushing for transparency and independent checks. This will help our innovations build trust, not just at home, but on the global stage.

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